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Hydroxyl group functionalized graphene oxide nanosheets as additive for improved erythritol latent heat storage performance: A comprehensive evaluation on the benefits and challenges

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ABSTRACT

Graphene oxide (GO) nanosheets were employed as the additive to make composites of erythritol, a promising medium-temperature PCM candidate. GO nanosheets modified with hydroxyl groups were applied to improve the dispersion stability of the composites. A systematic characterization on the latent heat storage performance was performed for both pure and composite erythritol, in order to identify the benefits and challenges of the composites. It was found that the thermal conductivity is increased by nearly twice and the degree of supercooling was lowered from ~64 °C to ~48 °C at the loading of 1.0 wt% GO nanosheets (the maximum loading tested). The addition of GO nanosheets also leads to an increase of the retrievable latent heat during crystallization, from ~187 kJ/kg to ~225 kJ/kg at the same loading, by increasing the crystallinity. However, the introduction of GO nanosheets can also lead to a rise in the dynamic viscosity of erythritol. As a result, the crystallization rate is slowed down and accordingly, the duration of crystallization becomes 62% longer when the loading reaches 1.0 wt%. In addition, favorable dispersion stability of the erythritol composites is observed, and their melting point (~117 °C) remains almost unchanged during 50 melting-crystallization cycles. Functionalized GO nanosheets have been shown to be an efficient additive for improving the performance of erythritol, but a trade-off analysis on the loading would be required to achieve the best overall performance.

1. Introduction

To improve the thermal efficiency of both industrial processes and residential applications, latent heat storage technology using solid-liquid phase change materials (PCMs) has been proposed as one of the promising thermal energy storage techniques due to its higher energy storage density and smaller temperature span during heat storage/retrieval processes, in comparison to sensible heat storage [1]. Latent heat storage has long been investigated and practiced in several fields like solar thermal utilization [2], industrial waste heat recovery [3], thermal management of electronic devices [4] and energy-saving buildings [5].

The primary requirement for latent heat storage applications is the exploration and identification of suitable PCMs at given temperature range. Although there are no universal standards on selection of PCM candidates so far, the following attributes, i.e., suitable melting point, high latent heat of phase change, high thermal conductivity, negligible supercooling effect, adequate rate of both nucleation and crystallization and favorable melt/crystallization cyclic stability in long-term operation are always basically required for a potential PCM candidate [6–8].

Taking erythritol (T_m = 119 °C), one of the familiar four-carbon-atom sugar alcohols as an example, it has been widely proposed as a potential PCM candidate for latent heat storage with temperature ~120 °C [9–11]. As the erythritol has much higher latent heat of fusion (~340 kJ/kg) in comparison to those of other PCM candidates with close melting point, such as the inorganic $MgCl_2 \cdot H_2O$ (T_m = 117 °C) with latent heat of fusion 169 kJ/kg [2,7] and the organic acetanilide (T_m = 118.9 °C) with latent heat of fusion 222 kJ/kg [8]. Furthermore, erythritol has relatively high thermal conductivity (~0.76 W/m °C at 27

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Nomenclature

c_p	specific heat, kJ/kg °C
ΔG	nucleation barrier, J
ΔH	latent heat, kJ/kg
k	thermal conductivity, W/m °C
m	mass, kg
ΔT	temperature difference, °C
T	temperature, °C
Δt	duration, min
u	standard uncertainty
V	volume, m^3
x	result of a given measurement
$\bar{x}$	arithmetic mean of results

Greek symbols

α	thermal diffusivity, m^2/s
γ	specific interfacial free energy, J
η	dynamic viscosity, Pa s
$\Delta\mu$	chemical potential difference, J/mol
ρ	density, kg/m^3
ω	mass fraction, %
Ω	volume of growth unit, mm^3

Subscripts

c	crystallization
cf	crystal and the mother phase
com	composite
ery	erythritol
het	heterogeneous
hom	homogeneous
i	a given measurement,
l	liquid
m	melting
n	number of measurements
p	pressure
s	solid
sc	supercooling

°C) [9], which is also higher than that of common organic PCMs like paraffin wax (k = ~0.2 W/m °C) [1]. Nevertheless, the thermal conductivity of erythritol can still be improved, and adding thermally enhanced nanomaterials has been identified as an effective approach [12–19]. For instance, it was experimentally found that the thermal conductivity of pure erythritol can be remarkably increased about 6.4 times (from 0.733 W/m °C to 4.72 W/m °C) in presence of 15 vol% expanded graphite [15].

In spite of the aforementioned desirable properties for PCM candidates, numbers of previous experimental investigations have found that the erythritol melt suffer from severe supercooling during both non-isothermal and isothermal cool-down process [20–28]. Actually, the supercooling behavior becomes one of the inherent characteristics for erythritol, and also for the other sugar alcohols. Because as polyalcohols, the reformation of the hydrogen bonds during cool-down takes place at a lower energy than the breaking of these bonds during the heating transition [29]. In addition to the experimental results, a recent review literature also confirmed the severe supercooling behaviors of sugar alcohols on the basis of an overall comparison of the main properties between sugar alcohols and other familiar PCM candidates like salt hydrates [30].

It is no doubt the supercooling effect should be suppressed, or even eliminated, as supercooling effect can really bring about some drawbacks against the practical applications. Specifically, it impedes the withdrawal of stored heat at the desired time. The supercooled PCM is out of thermodynamic equilibrium. The latent heat of crystallization may be lower than the latent heat of fusion because part of the heat is used by the supercooled PCM to reach the equilibrium temperature [31]. Our experimental results showed that the erythritol can only releases ~56% of latent heat of fusion upon crystallization as measured by DSC at a representative cooling rate of 5 °C/min [21]. In addition, the crystallization of a PCM candidate with supercooling behavior is less likely to take place at the required temperature range controlled by the system. It makes a system difficult to control and also result in additional energy consumption [32,33], especially for the low-temperature thermal energy storage systems like energy-saving buildings.

It has been confirmed that introducing suitable nucleating agents is an effective approach to suppress the supercooling of PCM candidates [25,26,34]. Based on the classical nucleation theory, the main physical mechanisms can attribute to the reduced free energy barrier of heterogeneous nucleation with nucleating agent, in comparison to homogeneous nucleation without nucleating agent [35,36]. The previous experimental results showed that the degree of supercooling of erythritol can be significantly lowered from 103.5 °C to 47.1 °C as measured by differential scanning calorimetry (DSC) at a cooling rate of 10 °C/min through introducing 1 wt% commercially available nucleating agent HPN-68 [25]. In addition to the commercial additives, hydroxyl group functionalized GO nanosheets were also used to mitigate the degree of supercooling, and improve simultaneously the thermal conductivity [37].

Attention should be paid that besides the thermal conductivity and supercooling, the melting point, latent heat of phase change, specific heat capacity, density, volume expansion, thermal stability, cyclic stability, corrosion resistance to container materials, and cost are also deemed to be important criteria for PCM selection [38]. It can be predicted that degradations on other aspects may arise along with the enhanced thermal conductivity and suppressed supercooling effect in the presence of the additives. However, based on a careful literature review, it was found that there is usually a lack of a comprehensive evaluation of the overall performance of the composites. Although the thermal conductivity of erythritol composites can increase to as high as ~30 W/m °C [18] and ~9 W/m °C [15] by adding percolated aluminum filler and spherical graphite, respectively, the loading has reached up to approximate 40 vol% without consideration of sacrifice in latent heat capacity and degradation of other transport properties, like increase in dynamic viscosity [39,40]. In addition to the phase change behaviors and transport properties, the compatibility between additives and pure PCM and the dispersion stability are also of primary importance for composite PCM. They are indispensably required to ensure the reliability of the measured properties and long-term operation. However, it was found that the dispersion stability test of composite PCM was done in very few experimental works [14,41]. Seemingly, the aforementioned HPN-68 can serve as potential nucleating agent of erythritol, but it appears to be uncertain about its reliability for long-term operation, as the compatibility or dispersion stability was not tested in the experiments [25].

Therefore, the present investigation proposed the thermally conductive graphene oxide (GO) nanosheets as a potential nucleating agent to suppress the supercooling of erythritol and simultaneously, improve its thermal conductivity. In consideration of the inherent polyhydric property of erythritol, the dispersion stability of the erythritol composite would be hopefully improved through using hydroxyl group functionalized GO nanosheets. In addition to the generated benefits, as the main interests of this work, a systematic characterization on the phase change behaviors and transport properties will be performed to clarify the possible challenges as a result of additive employment.

Furthermore, the cyclic stability of the erythritol composite was evaluated by conducting 50 consecutive melting-crystallization cycles under isothermal heating and cooling conditions. Based on the experimental results, suggestions would be provided to make the generated benefits outweigh the accompanied challenges to achieve better overall performance in real-world latent heat storage applications as much as possible.

2. Experimental

2.1. Materials

The specific procedures of the experimental work, which mainly contains preparation of erythritol composites, characterization on both the GO nanosheets and the erythritol composites, as well as phase change behaviors and transport properties tests were displayed systematically in Fig. 1. The GO nanosheets employed in this work are commercially available products (GaoxiTech, China) and its main specifications are listed in Table 1. As it becomes difficult to clearly observe the original micro-morphology of GO nanosheets after being dispersed in molten erythritol. The TEM characterization was performed using the GO-ethyl alcohol suspension with extremely low concentration of 0.01 wt%, and subsequently subjected to ultrasonication process for 30min to help the GO nanosheets disperse well. The TEM photograph was captured after the ethyl alcohol on the copper mesh volatilized completely. It can be clearly observed on the TEM microphotograph (Fig. 2) that the GO nanosheets are still likely to aggregate layer by layer, especially for the center area of the select scope. From the right picture with larger magnification we can find that the GO nanosheets present slightly wrinkled surface textures with curling edges.

Table 1
Specifications of GO nanosheets provided by the manufacturer.

Diameter (μm)	Thickness (nm)	Specific surface area (m^2/g)	Density (kg/m^3)	Purity (%)
1–10	2–8	10–300	10–1000	99

Meanwhile, the GO nanosheets were subjected to Fourier Transform Infrared (FTIR) spectroscopy and powder X-ray diffraction (XRD) tests. To be specific, the FTIR test was carried out on the Fourier Transform Infrared Spectrometer (FTIR, Nicolet5700, Thermo, USA) with infrared spectrum range 4000–400 cm^{-1} and resolution above 0.09 cm^{-1} at the room temperature (~25 °C). The powder XRD test was conducted on the X-ray diffraction (X-pert Powder, PANalytical B.V., Netherlands), also at the room temperature (~25 °C). The Cu-$K\alpha_{1,2}$ radiation source with a wave length of 0.1540598 nm was adopted. The XRD patterns were recorded at a step size of 0.02° in the range of 2–80°.

The erythritol reagent with purity 99% was provided by Energy Chemical (China). During the preparation of erythritol composites, a given amount of fresh erythritol powder was placed in a glass beaker. Then the glass beaker was immersed in a constant-temperature sand bath, which was heated from the bottom by a heating plate (IKA, HS7 control, German) with an accuracy ±1.0 °C. The erythritol powder was heated at a constant temperature of 139 °C (20.0 °C above its melting point) till it melted completely. Then a small amount of GO nanosheets powder with various mass fractions (0.01 wt%, 0.1 wt%, 0.5 wt%, 1.0 wt %) was added in the erythritol melt. Subsequently, the composites were

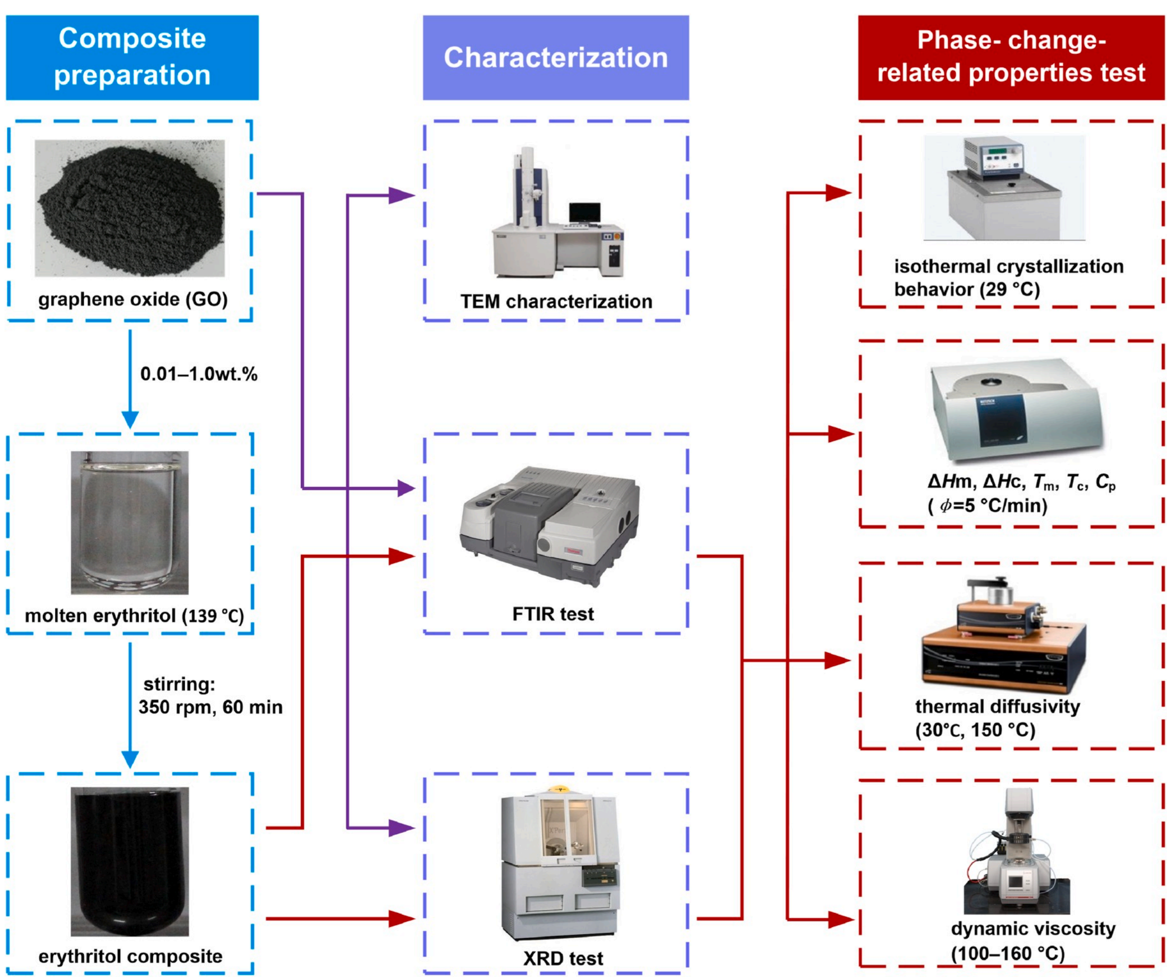

Fig. 1. Flowchart of the preparation, characterization and phase change performance tests for erythritol composites with GO nanosheets.

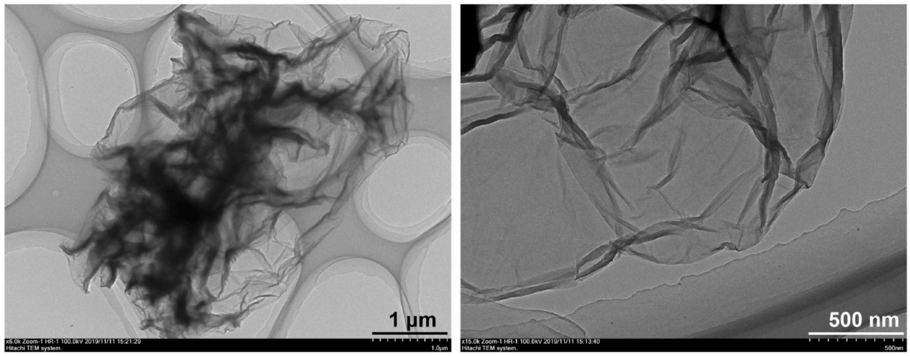

Fig. 2. TEM images showing the morphology of the GO nanosheets.

subjected to magnetic stirring at 500 rpm for 60 min and ultrasonic vibration with power of 650 W for 30 min, respectively to make GO nanosheets dispersed homogeneously in the erythritol melt. The two above-mentioned processes were carried out on the heat plate (IKA, HS7 control, German) and ultrasonic cell pulverizer (SCIENTZ, JY92-IIN), respectively. Afterwards, a small amount of the composite sample were taken out and solidified at the temperature of ~29 °C for the FTIR and XRD tests, whose corresponding procedures are as same as those of GO nanosheets. The rest molten sample was equally distributed into several smooth quartz glass test tubes (with length 100 mm, outer diameter 36 mm and inner diameter 30 mm) for the following tests.

2.2. Crystallization behavior tests

The specific procedures on the crystallization behaviors tests have been introduced in our previous report [23]. It should be emphasized that the mass of the pure erythritol melt is 5 g in each individual sample. In consideration of the practical latent heat retrieval condition with temperature around environment temperature, the cooling temperature in the present study were set at a constant temperature of 29 °C. The transient temperature history during crystallization process was recorded by an embedded J-type thermocouple (with calibrated accuracy of ±0.5 °C) along the vertical axis of the test tube. The bead of the thermocouple was suspended 5 mm above the inner bottom of the test tube. The data acquisition was accomplished by a computer-based data logger system. Prior to the test, the temperature distribution of the pure erythritol sample was verified by using three thermocouples simultaneously along the radial and vertical directions, respectively.

2.3. Transport property tests

The latent heats of phase change and key transport properties, including dynamic viscosities, specific heat capacity, thermal diffusivity, density and thermal conductivities were characterized for the pure and composite erythritol samples to provide an overall evaluation on the performance of erythritol composites. The characterization of latent heats of fusion and crystallization were carried out on the differential scanning calorimeter (DSC, Netzsch DSC 200F3) at a representative ramping/cooling rate of 5 °C/min. Three consecutive melting-crystallization cycles were tested after eliminating thermal history and the DSC curves in the first cycle were presented. The dynamic viscosity tests were conducted on a stress-controlled modular rheometer (Anton Paar, MCR102) at a broad shear rate range (0.1–100/s) for the sample at both molten and supercooled liquid states [42]. The thermal diffusivities of the samples in solid state (30 °C) and molten state (150 °C) were measured on a Flash Diffusivity Analyzer (TA, DXF-200). The densities ρ of the samples, also in both solid (30 °C) and molten states (150 °C), were determined by means of measuring their respective masses m as well as volumes V [9], and calculated according to equation (1),

$$\rho = \frac{m}{V} \tag{1}$$

The specific heat capacity as a function of temperature was characterized from a broad temperature range (30–150 °C) under same ramping rate 5 °C/min using the same DSC. Thus the thermal conductivities k (W/m °C) of the samples can be determined with the aid of their respective thermal diffusivities α (m^2/s), densities ρ (kg/m^3) and specific heat capacities c_p (kJ/kg °C) at given temperatures based on equation (2) [13,14,19,43,44],

$$k = \alpha \rho c_p \tag{2}$$

2.4. Uncertainty analysis

Parallel tests were performed on six individual specimens of each sample to ensure data reproducibility. The exact average values and their corresponding standard uncertainties were documented for each case. The uncertainty analysis was performed for all of the measured results based on the type A standard uncertainty [12,45,46], which follows,

$$u(x_i) = \sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \Big/ n-1}, \tag{3}$$

where $u(x_i)$ means the standard uncertainty of a single measurement, n is the number of the measurements, x_i is the result of a given measurement, $\bar{x}$ is the arithmetic mean of all measurement results.

3. Results and discussion

3.1. Characterization results

The FTIR spectra are used to confirm the chemical functional groups of GO nanosheets, pure erythritol and erythritol composites. The characteristic absorbance peaks locating at 3392.4 cm^{-1} and 1385.2 cm^{-1} correspond to the stretching vibration and bending vibration, respectively of hydroxyl group –OH on the surface of GO nanosheets [37,47], indicating the existence of –OH (Fig. 3(a)). In addition, the appearance of the rest two obvious peaks at 1647.1 cm^{-1} and 1110 cm^{-1} attribute to the stretching vibrations of C=C and C–O functional groups, respectively. For the FTIR spectrum of the pure erythritol, it is found to be in accordance with that in several previous studies [19,37,41,47]. It should be emphasized that the strong absorbance peak at 3250.6 cm^{-1} denotes the stretching vibration of –OH functional group. The result indicates the inherent polyhydric property of erythritol. In comparison to the GO nanosheets and pure erythritol, the FTIR spectrum of the erythritol composite stays similar to that of pure erythritol in the range of 3000–400 cm^{-1}. However, three new sharp and narrow absorbance peaks were achieved at the wavenumbers around 3300 cm^{-1}. The two

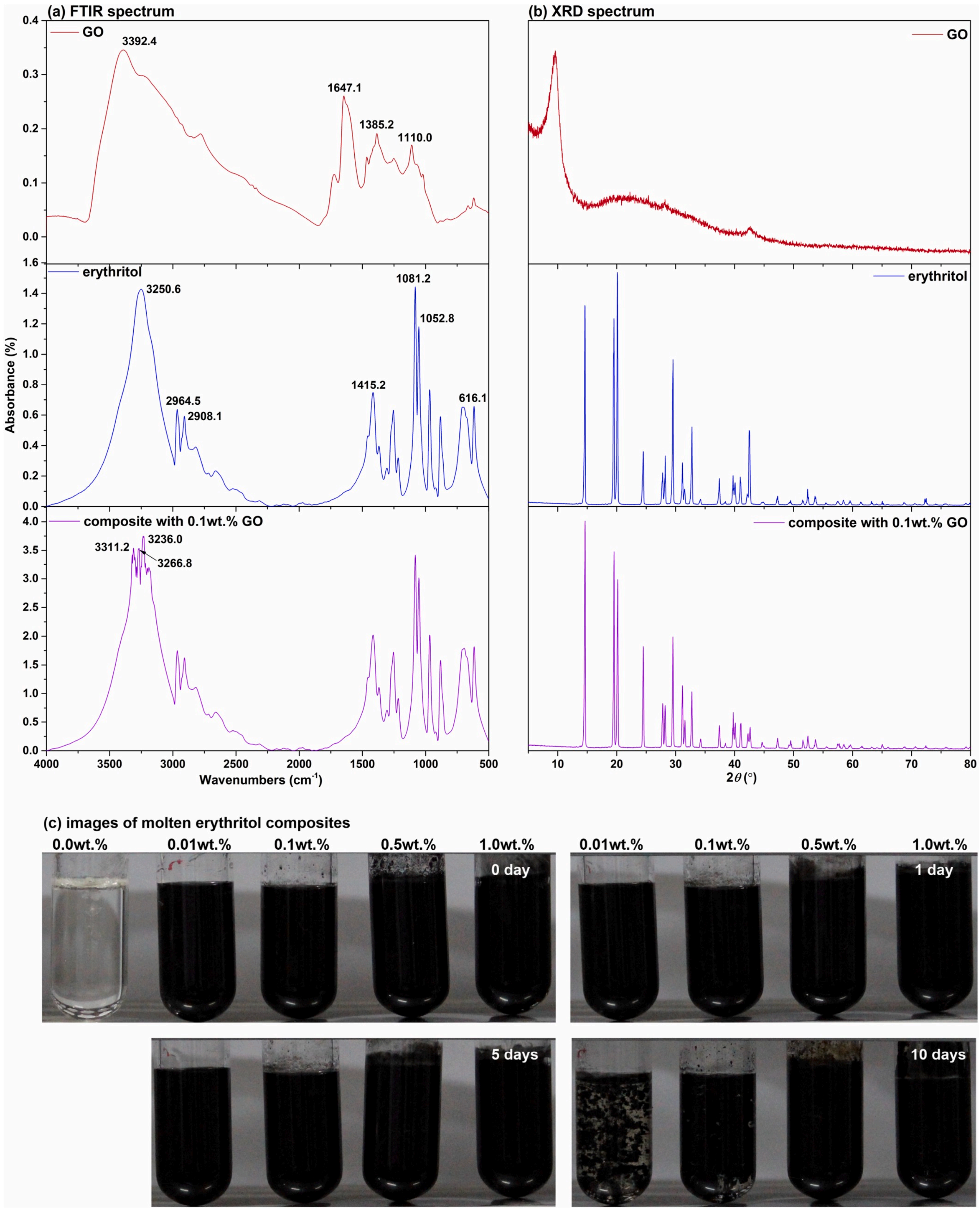

Fig. 3. (a) FT-IR spectra, (b) XRD patterns and (c) photographs of erythritol and its composites.

main peaks at 3311.2 cm^{-1} and 3236.0 cm^{-1}, also corresponding to the stretching vibration of –OH functional groups shift to lower wavenumbers compared to those of GO nanosheets and erythritol. The results imply the formation of hydrogen bonds in the composites due to the hydroxyl groups. The rest peak at 3266.8 cm^{-1} also represents the hydrogen bonds. Similar results were also found in polyethylene glycol-GO composites [48].

The XRD tests were conducted to further confirm whether chemical reactions take place during the mixing process. It can be clearly observed that the main diffraction peak of GO nanosheets lies in the position with 2θ value of ~10° (Fig. 3(b)). For the erythritol, the main diffraction peaks locate at 14.6°, 19.5°, 20.1°, 24.5°, 29.5°, 32.8° and 42.6°, which appears to be similar to that of the erythritol. The absence of new diffraction peaks indicate that there are no chemical reactions happened during the preparation of the erythritol composites, and they were just mixed physically. Attention should be paid that the diffraction peak of the GO nanosheets disappears on the XRD spectrum of the composite. In addition to the very low loading (0.1 wt%) of GO nanosheets, the above result was possibly caused by exfoliation and unordered structure of GO nanosheets after dispersing in the molten erythritol [37,49,50]. The analogous results were also found at relatively higher loadings (1 wt%) of GO [37,49].

Some methods like ultraviolet–visible (UV–Vis) spectrophotometer measurements, light scattering method and sediment photography, etc., have been widely used to estimate the relative dispersion stability of nanofluids and composite PCMs in liquid state. The UV–Vis spectrophotometer and light scattering method can generate stability results in a quantitative manner [51]. However, the temperature of the commercial instruments are unable to reach as high as 130 °C for liquid erythritol. Therefore, the sediment photography method was used to roughly evaluate the dispersion stability of the GO-erythritol composites. The same method was also adopted in TiO_2-erythritol composites [41]. Based on the images captured for the fresh erythritol composites and the samples kept as long as for 10 days at 130 °C (Fig. 3 (c)), it can be clearly observed that the appearance of molten erythritol composites still remain almost the same after 1 day and 5 days. By contrast, the composite with minimum mass fraction 0.01 wt% showed some clusters as a result of the aggregation of GO nanosheets. Fortunately, the clusters were able to stay suspended in the molten erythritol without significant settling down to the bottom of the glass tube. Similar result is also obtained for the sample with 0.1 wt% GO nanosheets, i.e., a few clusters can be found after 10 days. By contrast, the clusters can be hardly seen for the samples with higher mass fraction (0.5 wt% and 1.0 wt%) of GO nanosheets, and still remain similar appearance to the fresh ones. As the density of GO nanosheets lies in the range of 10–1000 kg/m^3, which is lower than that of liquid erythritol (~1300 kg/m^3). Furthermore, the absence of sedimentation of GO nanosheets also attributes to the higher dynamic viscosity of liquid erythritol, in comparison to that of other organic PCMs like paraffins. For instance, it was experimentally found that the erythritol present a dynamic viscosity of approximate 5 times higher than that of paraffin at the temperature of about 10 °C above their melting points [42,52]. There was no apparent sedimentation after 10 days. This observation suggested an acceptable stability of the composites, thanks to the favorable compatibility between the erythritol and the functionalized GO nanosheets, the high dynamic viscosity of liquid erythritol, as well as the low density of GO nanosheets.

3.2. Phase change behaviors

The temperature distributions of the pure erythritol sample along the radial and vertical directions are displayed in Fig. 4(a) and (b) respectively. It was found that the uniform temperature distributions along the two directions is achieved before the initiation and after the crystallization. In contrast, slight temperature difference appears during the process of crystallization, where solid and liquid phases coexist. The main reasons for the temperature difference is attributed to the randomness of crystallization process and the degraded performance of heat transfer.

In consideration of the inherent randomness in crystallization process, *T*-history of pure erythritol (Fig. 5(a)) and erythritol composites (Fig. 5(b)–(e)) were performed for six individual specimens to verify the data reproducibility. Slight temperature difference (<5 °C) appeared during crystallization process, especially for pure erythritol sample 5. Similar results were also found for composite sample 2 with 0.01 wt% GO nanosheets (Fig. 5(b)), composite sample 2 with 0.1 wt% GO nanosheets (Fig. 5(c)), composite sample 4 with 0.5 wt% GO nanosheets (Fig. 5(d)) and samples 2, 3 of composite with 1.0 wt% GO nanosheets (Fig. 5(e)). In reality, it seems to be a challenge to obtain completely identical experimental results for most of the composites. Hence, it is of necessity to conduct the tests using parallel individual samples as many as possible, and the results with large deviation should be extracted out during the subsequent process of results analysis.

A detailed analysis on the *T*-history curve of erythritol has been presented in our previous study [23]. Hence, emphasis will be put on the composites. The *T*-history curves of erythritol composites are in similar pattern to that of the pure erythritol. In the case of a typical *T*-history curve of erythritol composite sample 1 with 0.01 wt% GO nanosheets (Fig. 5(b)), the temperature first gradually declines to a temperature far below its melting point (117.3 °C), which refers to its nucleation (or crystallization) temperature (61.3 °C). Once nuclei start to form, the crystallization initiates along with retrieval of latent heat of fusion. During the initial stage of crystallization, the heat generation is much

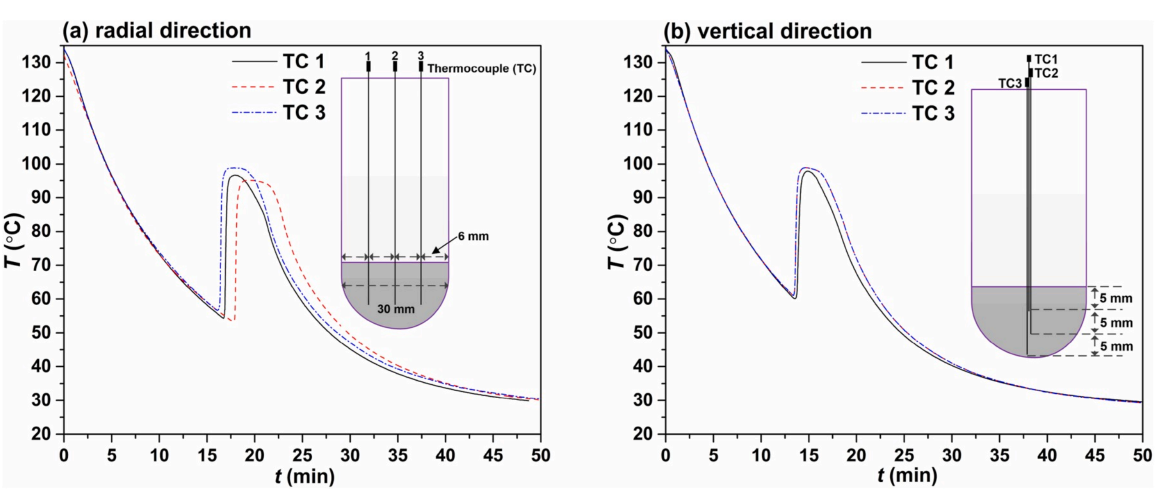

Fig. 4. Temperature distributions inside the erythritol sample along the (a) radial and (b) vertical directions.

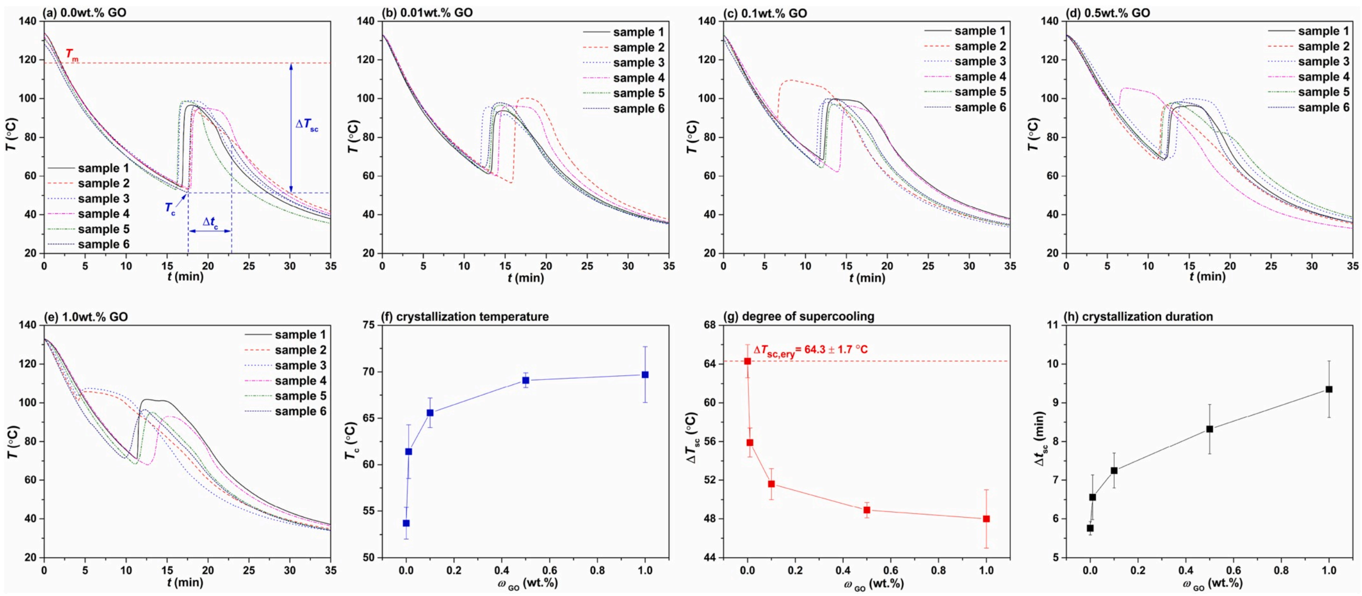

Fig. 5. *T*-history curves of (a) pure erythritol, and erythritol composite with (b) 0.01 wt%, (c) 0.1 wt%, (d) 0.5 wt%, and (e) 1.0 wt% GO nanosheets, and (f) crystallization temperature, (g) degree of supercooling, and (h) duration of crystallization as a function of the loading of GO nanosheets.

faster than the heat transfer towards the surroundings. Therefore, the temperature dramatically rise to a maximum value of 93.8 °C, followed by a levelled-off period at around this temperature, which indicates that the heat generation rate becomes nearly equal to that of heat transfer during the period. With the crystallization continuing to move forward, the temperature of the sample tend to decline, meaning that the rate of heat generation is much slower than that of heat transportation. The results demonstrate that the crystallization progress is approaching the end. The completion of the crystallization process can be indicated by a transition point where the shape of the *T*-history curve changes from up-convex to down-concave. The entire duration of crystallization lasts about 6.8 min. Afterwards, the temperature of the sample declined more slowly until it reached the cooling temperature (29 °C) in the end. The *T*-history curves of the rest composites with 0.1 wt%, 0.5 wt% and 1.0 wt % GO nanosheets show similar pattern to that of the composite with 0.01 wt% GO nanosheets. The crystallization temperature, degree of supercooling and crystallization duration of the crystallization are exemplified in Fig. 5 (a).

The crystallization temperatures T_c of erythritol composites increase with the mass fraction of GO nanosheets (Fig. 5(f)). A sharp increase in crystallization temperature is obtained with mass fraction of GO at 0.1 wt%. The crystallization temperature tends to remain when the mass fraction approaches 1.0 wt%. Accordingly, the degree of supercooling ΔT_{sc} can be determined by the temperature difference between the experimentally determined melting point T_m and crystallization temperature T_c [30–33] (Fig. 5 (a)). It is apparent that a high crystallization temperature corresponds to a low degree of supercooling ΔT_{sc}, as it was experimentally found that the introduction of GO nanosheets have negligible influence on the melting point of pure erythritol (Fig. 5(g)). A more precise measurement on the melting points of the erythritol composite was carried out with DSC. An extremely small amount of GO nanosheets (0.01 wt%) can lower the degree of supercooling of erythritol from ~64 °C to ~56 °C. The degree of supercooling of erythritol can be further reduced with continuous increase in mass fraction of GO nanosheets. A minimum degree of supercooling (48 °C) is obtained for the composite with 1.0 wt% GO nanosheets, which corresponds to a reduction of 16 °C compared to that of pure erythritol (64 °C). The results demonstrate that the GO nanosheets can really serve as potential nucleating agent to suppress the supercooling behavior of erythritol.

The physical mechanisms on the above results can be analyzed through the classical nucleation theory [35]. In the process of homogeneous nucleation, an embryo in increasing its size must overcome a free energy barrier, i.e., a homogeneous nucleation barrier $\Delta G^{*}_{\text{hom}}$ before it develops a stable crystal, which reads:

$$\Delta G^{*}_{\text{hom}} = \frac{16\pi}{3}\frac{\Omega^2\gamma_{\text{cf}}^3}{\Delta\mu^2} \tag{4}$$

Where Ω is the volume of growth unit, γ_{cf} is the specific interfacial free energy between the crystal and the mother phase, $\Delta\mu$ is the chemical potential difference between the actual state and the equilibrium state. After introducing the foreign particles, the foreign particles reduce to embryos of the crystallizing phase and the nucleation becomes heterogeneous nucleation [36]. The nucleation barrier in the case of heterogeneous nucleation can be calculated with Equation (5):

$$\Delta G^{*}_{\text{het}} = \Delta G^{*}_{\text{hom}} f(m, x) \tag{5}$$

The factor $f(m, x)$ quantifies the reduction of the nucleation barrier from homogeneous nucleation to heterogeneous nucleation due to the occurrence of foreign bodies, ranging from 0 to 1. This explains why the nucleation agents are used to suppress supercooling. However, the GO nanosheets tend to aggregate at higher concentration, which means that the effective nucleating sites will level out. Therefore, the degree of supercooling suppression is limited after a specific amount of GO nanosheets.

In parallel, the crystallization time becomes longer after introducing nucleating agent GO nanosheets (Fig. 5(h)). Even with low mass fraction of GO nanosheets (0.01 wt%), it can also lead to a certain increase in crystallization duration of erythritol (from ~5.76min to 6.56min). With mass fraction of GO nanosheets further increase from 0.1 wt% to 1.0 wt %, the crystallization duration almost increase linearly from ~7.25 min to ~9.35 min. It is easy to understand that a longer crystallization duration corresponds to a slower rate of crystal growth rate. The previous studies stated that the propensity of the supercooled liquid to crystallize is essentially determined by the molecular mobility and the thermodynamic driving force [53]. A high dynamic viscosity limits the molecule mobility and it will become difficult for molecules to undertake crystal formation [54–56], leading thus to a slower rate of crystal growth. The dynamic viscosity in supercooled liquid state plays important role in the crystallization process.

The dynamic viscosity as a function of mass fraction of GO nanosheets was measured for erythritol composites in both molten and supercooled states. The dynamic viscosity increases even with small amount of GO nanosheets (0.01 wt%) (Fig. 6). The dynamic viscosity increases linearly from 0.051 Pa to 0.057 Pa s with mass fraction of GO nanosheets increase from 0.1 wt% to 1.0 wt% in supercooled liquid state (108.8 °C). As a result, the crystal growth slows, thus the crystallization durations extend and the rate of latent heat release is reduced. The dynamic viscosities increase approximate 40% in presence of 1.0 wt% GO nanosheets in the molten state (158.8 °C). The dynamic viscosity increase in molten state is unfavorable as it deteriorates the natural convection during melting and results in a slower storage rate.

The phase change behaviors of pure erythritol and its composites were also tested in non-isothermal conditions to determine the latent heat of phase change. The DSC thermograms of the samples are presented in Fig. 7(a)–(e). The melting point of erythritol remained at around 119 °C and the crystallization temperatures improved after introducing GO nanosheets (Fig. 7(f)), which is in accordance with that tested in isothermal method. Accordingly, the degrees of supercooling of erythritol were significantly reduced from 104 °C to 66 °C in presence of 1.0 wt% GO nanosheets (Fig. 7(g)), compared to a lower reduction from 90.3 °C to 73.4 °C with more GO (3.0 wt%) at same cooling rate of 5 °C/min [37]. The results indicated that despite the similar hydroxyl group functionalized GO nanosheets, better performance in suppressing the supercooling was achieved in the present work. Interestingly, the degrees of supercooling measured by non-isothermal method appear to be always higher than that by isothermal method at each mass fraction of GO nanosheets. The results are closely related to the small size of sample (10–20 mg) in non-isothermal test, it tends to suffer from very serious supercooling.

As the additives are unable to contribute to the latent heat of fusion, total storage enthalpy of the storage will be reduced. The latent heat of fusion is reduced from 332 kJ/kg to 322 kJ/kg in presence of 0.01 wt% GO nanosheets. Interestingly, the reduction of latent heat of fusion tends to slow down with further increase in mass fraction of GO nanosheets from 0.1 wt% to 1.0 wt%. The similar trend was also obtained in Cu/paraffin composites [57]. It was theoretically proposed that the latent heats of fusion were expected to linearly decrease with additive loadings, and lots of experimental results were found to follow this trend [58]. However, remarkable disagreement between the theoretical prediction and measurement, as well as the non-linear trend were also reported as well [57]. In spite of the proposed effects like interfacial liquid layering, Brownian motion, and particle clustering [58], the underlying mechanism governing the reduction of latent heat of fusion has not yet

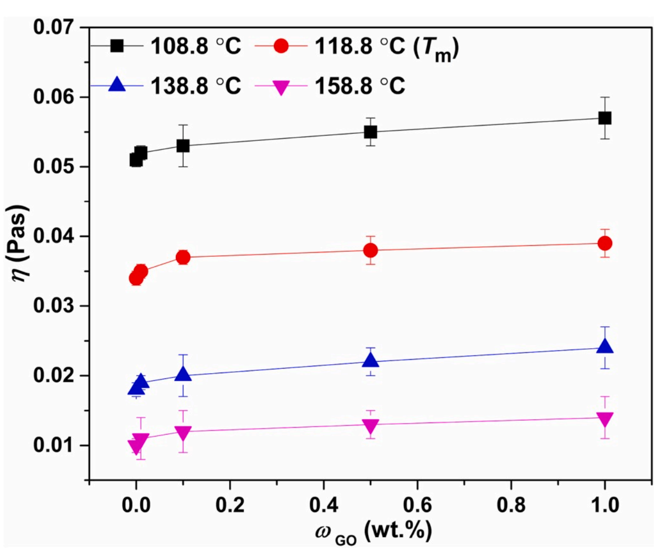

Fig. 6. Dynamic viscosity of erythritol composites as a function of the loading of GO nanosheets.

been fully understood as of now. It should be noted that, unlike common PCMs like alkanes and paraffin, the hydrogen bonds in sugar alcohols play an important role in their quantity of latent heat [10]. Therefore, it can be hypothesized that the hydrogen bonds in GO-erythritol composites lead to compensation of the latent heat sacrifice, and only 13.1% loss in latent heat storage capacity was observed when the loading of GO nanosheets reaches the maximum of 1.0 wt%. The mechanism on the latent heat of fusion could be further investigated for the composites with formation of hydrogen bonds.

Fortunately, the addition of GO nanosheets improves the latent heat of crystallization, rising from 186.8 kJ/kg to 224.8 kJ/kg in presence of 1.0 wt% GO nanosheets. The contribution of GO nanosheets to the latent heat of crystallization of erythritol is mainly attributed to the suppressed supercooling effect and improved crystallinity. The improvement of latent heat of crystallization with loading of GO nanosheets, i.e., the variation trend of a rapid increase followed by a slower section was found to be in a similar pattern to the variation of the degree of supercooling.

3.3. Transport properties

In addition to the dynamic viscosity, the other four important transport properties, i.e., thermal diffusivity, density, specific heat capacity and thermal conductivity were also measured for the erythritol composites in both solid (30 °C) and liquid (150 °C) states. The thermal diffusivity of erythritol can be greatly enhanced with addition of GO (Fig. 8(a)). The improvements reach as high as 154% and 78% for erythritol composites in solid and liquid states, respectively in presence of 1.0 wt% GO nanosheets. By contrast, the densities of erythritol composites tend to slightly decrease after introducing GO nanosheets, as the density of GO nanosheets (10–1000 kg/m^3) is lower than those of erythritol in both solid (1460 kg/m^3) and liquid (1292 kg/m^3) state (Fig. 8(b)). The volume expansion can be determined by $(\rho_s\text{-}\rho_l)/\rho_l$ accordingly. The variation of volume expansion of erythritol shows less than 1%, indicating the negligible influence of adding GO nanosheets.

Interestingly, the specific heat capacity of erythritol composite also get improved with addition of GO nanosheets (Fig. 8(c)). For the specific heat capacity of erythritol in liquid state, it first fast rises from 2.71 kJ/kg °C to 2.90 kJ/kg °C with mass fraction of GO increase to 0.1 wt%, and then slowly rise to 2.96 kJ/kg °C with 1.0 wt% GO nanosheets. Similar results were also obtained for the specific heat capacity in solid state. In the literature, it has been stated that the specific heat capacity of PCMs (like molten salts) after introducing additives showed some contradicting results. While some studies showed enhancement in specific heat capacity, others showed degradation. The explanations for these results are yet to be proposed [59]. A recent study reported a 45% enhancement in heat capacity of solid erythritol with 0.2 vol% of nano-titania, which was predominantly due to increased amount of interface [41]. Combining the above three properties, the thermal conductivities can be determined. The thermal conductivities are greatly improved in both solid and liquid states (Fig. 8(d)). The improvement of thermal conductivity was found to be 187% and 94% for the erythritol composites in solid and liquid states, respectively. Both are also more efficient than the results achieved in Refs. [37] with the same type of additive material.

In addition to the determination of the thermal conductivities, the enhancement of thermal conductivity of erythritol with additives in available reports is also reviewed. On account of the variation of mass or volume fraction in different studies, 1.0 wt% was adopted for comparison (Table 2). Although an enhancement as high as approximate four times by adding 1.0 wt% multi-walled carbon nanotubes was achieved for solid erythritol at 50 °C [14]. We are uncertain about the accuracy of the result, as the measured thermal conductivity of pure erythritol (0.196 W/m °C) appears to be much lower than most of the reported ones (~0.75 W/m °C). Apart from this exception, it can be found the thermal conductivity enhancement of the present work is still higher than the rest results in solid state. It is confirmed that the GO nanosheets

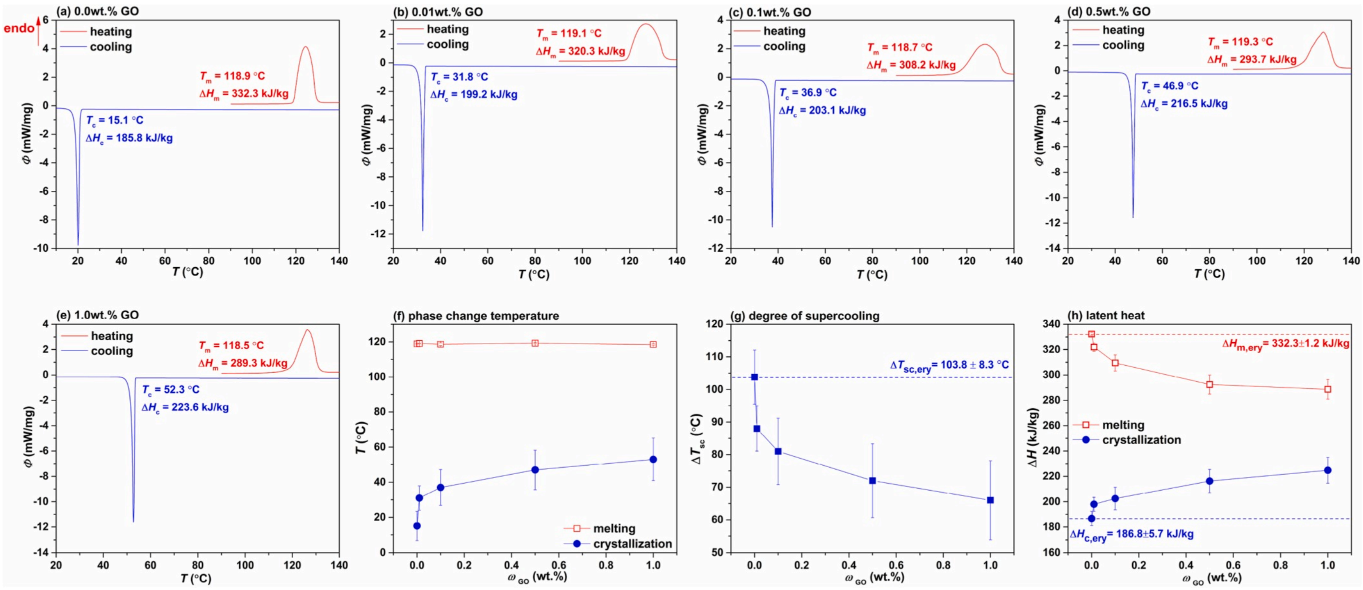

Fig. 7. DSC curves of (a) pure erythritol, and erythritol composite with (b) 0.01 wt%, (c) 0.1 wt%, (d) 0.5 wt%, and (e) 1.0 wt% GO nanosheets, (f) phase change temperatures, (g) degree of supercooling and (h) latent heat of phase change as a function of the loading of GO nanosheets.

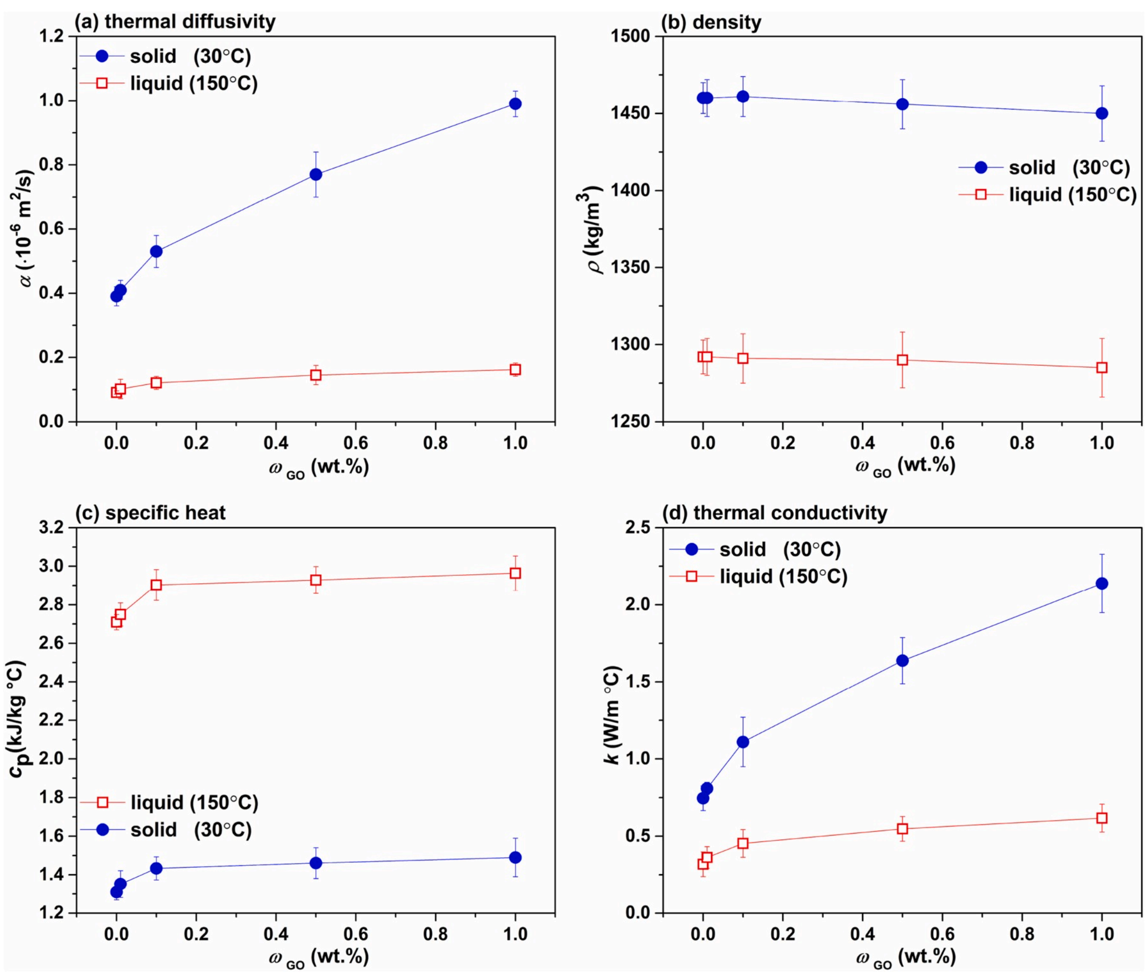

Fig. 8. (a) Thermal diffusivity, (b) density, (c) specific heat capacity, and (d) thermal conductivity of erythritol composite as a function of the loading of GO nanosheets.

Table 2
Comparison of the relative thermal conductivity enhancement of erythritol with various additives at the same loading of 1.0 wt%.

	Phase (T)	k_{ery} (W/m °C)	Additives	k_{com} (W/m °C)	Relative enhancement (%)	Measurement method
Present work	solid (30 °C)	0.75	GO nanosheets	2.14	185	laser flash method
	liquid (150 °C)	0.32		0.62	94	
Literature data	solid (–) [12]	0.41	expanded graphite	0.76	85	transient hot wire method
			carbon nanotubes	0.85	42	
	solid (25 °C) [13]	0.76	short carbon fibers	1.56	105	laser flash method
	solid (50 °C) [14]	0.20	multi-walled carbon nanotubes	0.98	400	transient plane source method
	solid (–) [19]	0.73	graphene nanoparticles	1.12	53	laser flash method

can serve as potential additive to improve the thermal conductivity of erythritol and suppress its supercooling. Attention should be paid that the above properties depend on the temperature. As the main objective of the present work is to evaluate the benefits and challenges of the composite PCM, the temperature-dependent behaviors of these properties were not pursued.

3.4. Cyclic stability

The cyclic stability of erythritol composite with 1.0 wt%, the maximum loading in the present work was tested in 50 isothermal melting-crystallization cycles. Based on the T-history curves of four representative cycles, i.e., the 1st, 15th, 30th and 50th cycle (Fig. 9(a)), it was found that the T-history curves in melting process approximately overlap. In contrast, a slight temperature difference was observed in the crystallization process due to the randomness of crystallization initiation. Furthermore, a more specific evaluation on the cyclic stability was performed by the variation of melting and crystallization temperatures as a function of cycles. It can be intuitively observed that the melting point remained at around 117 °C during the 50 cycles, and the crystallization temperature showed some variation in the range of 63.2–71.4 °C. In view of the randomness property of crystallization process, which was found in the crystallization of pure erythritol, the erythritol composite shows relatively good cyclic stability. Further melting-crystallization cycling test is recommended for long term erythritol composite use as PCM.

For the rest two properties, thermal stability and corrosion resistance, it has been found that the introduction of additives shows negligible effect on the thermal stability of pure PCM based on the TGA test [60]. The corrosion of sugar alcohols to common metals (aluminum, stainless steel, carbon steel and copper) was identified to be insignificant and the corrosion rates were determined to be less than 0.2 mm per year [61]. Therefore, the corrosion test was not considered in the present work.

3.5. A comprehensive evaluation

Based on the experimental results, a systematic summary on the generated benefits and challenges for composite PCM after introducing GO additives is compiled in Fig. 10. The phase change behaviors and transport properties can be improved at different level with GO nanosheets. However, some challenges such as the extension of crystallization duration also occur, which is confirmed to be inevitable. The cyclic stability is an important parameter and it seems difficult to accurately evaluate it before a large number of melting-crystallization cycles. In addition to the phase change behaviors, transport properties and cyclic stability, the cost is also one of the important factors in PCM selection. To the best of the authors' knowledge, introducing additives may possibly increase the cost, especially like the GO nanosheets, an additive at 500 times higher cost than that of erythritol employed in the present study. Nevertheless, the cost of commercially available GO nanosheets (or other additives) strongly depends on their quality, manufacturer and other factors.

Particular latent heat storage applications require specific improvements of the properties of the PCM. Therefore, it is preferable to evaluate the performance of composite PCMs under specific heat storage/

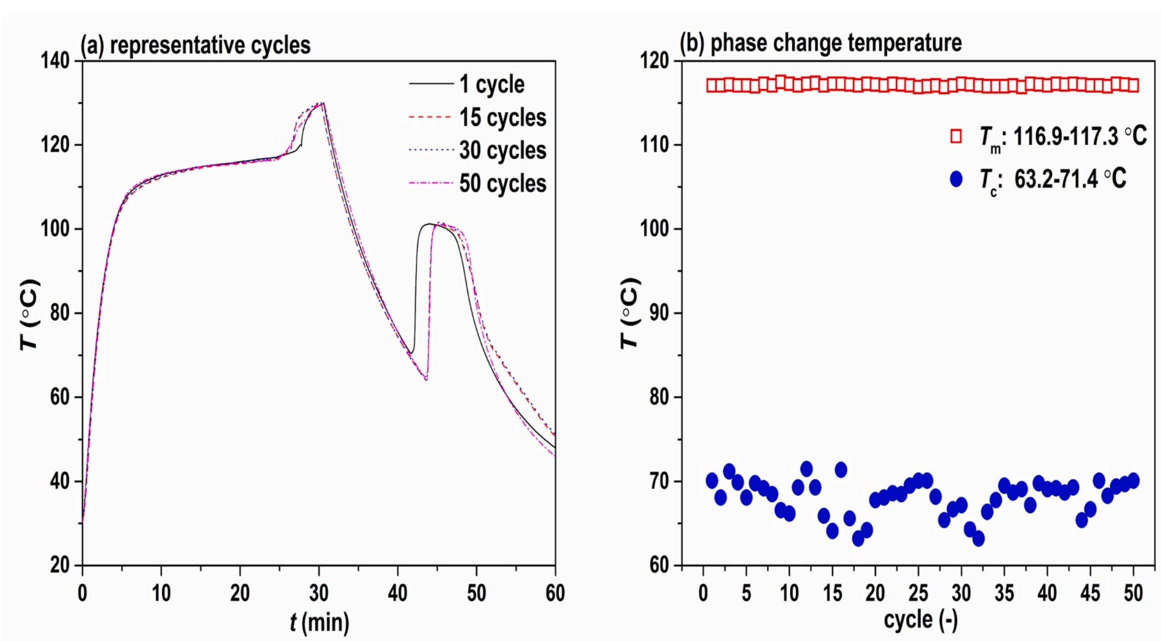

Fig. 9. Cyclic stability of the erythritol composite with 1.0 wt% GO nanosheets: (a) representative cycles and (b) phase change temperature as a function of number of cycles.

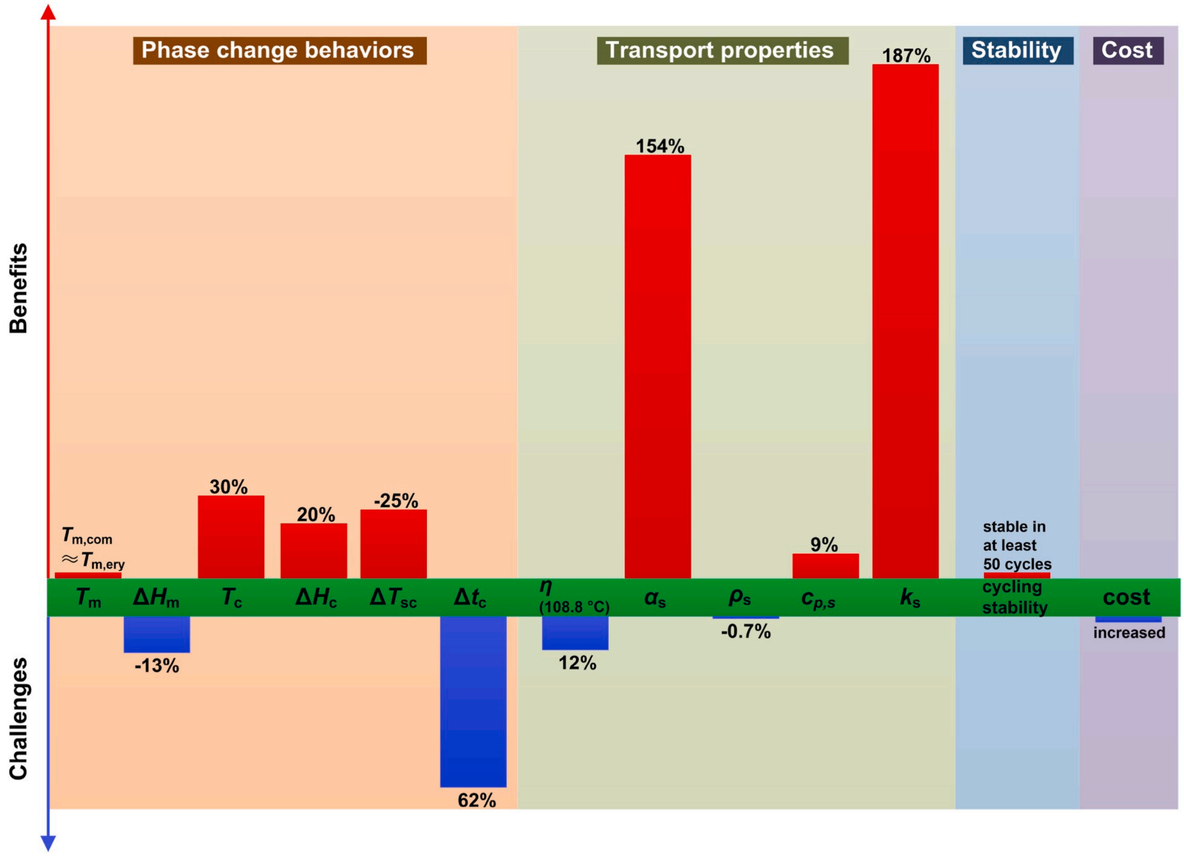

Fig. 10. A summary on the benefits and challenges for preparing erythritol composite with the addition of GO nanosheets (taking the case of 1.0 wt% loading as an example).

retrieval conditions. The present study suggests that both the benefits and challenges strongly depend on the loadings of additives. The better overall performance can be achieved by controlling the loadings of additives for the particular applications.

4. Conclusions

In order to clarify the resultant benefits and challenges after introducing additives into a PCM, a systematic characterization on the phase change behaviors and transport properties was performed for erythritol composite PCM with hydroxyl group functionalized GO nanosheets as the additive. The main conclusions can be drawn as follows:

(a) The molten erythritol composites are likely to stay stable and there is no apparent precipitation in the samples with higher mass fraction of GO nanosheets (0.5 wt% and 1.0 wt%) after 10 days. This compatibility between erythritol and GO nanosheets is attributed to the hydroxyl functional groups on its surface.

(b) The degree of supercooling is suppressed from 64.3 °C to 48 °C during isothermal crystallization process (29 °C) in presence of 1.0 wt% GO nanosheets, as a result of triggering heterogeneous nucleation. Interestingly, the addition of GO also leads to an increase of the retrievable latent heat during crystallization of erythritol, from ~187 kJ/kg to ~225 kJ/kg at the same GO loading of 1.0 wt%.

(c) Unfortunately, the addition of GO nanosheets brings about a 13% sacrifice on latent heat of fusion. In addition, it also slows down the crystallization rate of erythritol due to the increased dynamic viscosity, as indicated by an extension of the crystallization duration from 5.76 min to 9.35 min in presence of 1.0 wt% GO nanosheets. Specifically, the dynamic viscosity of supercooled erythritol is increased by 12% at 108.8 °C.

(d) The addition of GO nanosheets improves the thermal diffusivity, specific heat capacity of erythritol and slightly reduces its density. A 185% enhancement of thermal conductivity is achieved for solid erythritol (30 °C) after introducing 1.0 wt% GO nanosheets, which is found to be higher than most of the reported ones with the same additive loading.

(e) The cyclic stability test demonstrates that the melting point remains unchanged (~117 °C) in 50 consecutive melting-crystallization cycles. However, the crystallization temperature shows variation in the range of 63.2–71.4 °C.

(f) It is suggested that the loadings of additives should be controlled in a suitable range to achieve better overall performance. Although a comprehensive evaluation was done by focusing on several important properties of the composite PCM in this work, it would never be a complete one. More properties could be appended to the evaluation criterion list, and some of which may only be able to be examined with lab- or pilot-scale tests, e.g., the actual heat storage/retrieval performance under specific application conditions.

Declaration of competing interest

On behalf of the co-authors of the manuscript "Hydroxyl group functionalized graphene oxide nanosheets as additive for improved erythritol latent heat storage performance: A comprehensive evaluation on the benefits and challenges" submitted for publication in Solar Energy Materials and Solar Cells, I hereby declare that there are no actual or potential conflict of interests including any financial, personal or other relationships with other people or organizations that could inappropriately influence, or be perceived to influence, our work.

CRediT authorship contribution statement

Xue-Feng Shao: Investigation, Data curation, Visualization, Writing - original draft. **Jia-Cheng Lin:** Investigation, Writing - original draft. **Hao-Ran Teng:** Investigation, Writing - original draft. **Sheng Yang:** Investigation. **Li-Wu Fan:** Conceptualization, Supervision, Project administration, Writing - review & editing. **Justin NingWei Chiu:** Supervision, Project administration, Writing - review & editing. **Zi-Tao Yu:** Supervision, Project administration. **Viktoria Martin:** Supervision, Project administration.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solmat.2020.110658.

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